RETRACTED AS PROOF

ARCHIVE HISTORIQUE - T1'-T4 argument invalidated

Editorial notice - 20 October 2026

The historical T1'-T4 chain does not prove the Riemann hypothesis. The public erratum of 13 October 2026 identifies errors in T2 (weighted Gram-matrix bound) and T3 (wave-packet normalization and capture argument).

Earlier titles, abstracts and claims of complete or journal-ready proofs are historical claims withdrawn as proof arguments. This withdrawal does not assert that every mathematical object cited is false.

This editorial page has been added to the historical PDF. All original pages follow with their content unchanged. This is neither a recompilation nor a new mathematical result.

Source document: docs/demo_T1p_T4_EN.pdf

Original PDF SHA-256:
`8cd920196af9768c053ee84b2ac6302c806e0144e7851811547997e3357dc92f`

Public T1'-T4 erratum - 13 October 2026

Demonstration note — T1′→T4 analytic scheme (non-resolutive)

Alexandre Couret
InterIA Mathematical Collective

February 27, 2026

Abstract

Caution. RH is open. This note demonstrates the *analytic scheme* T1′–T4: (T1′) explicit formula (Guinand–Weil), (T2) spectral framing (Gram + Gershgorin/Schur), (T3) wave-packet localization, (T4) cross-positivity. We prove *the bounds* and *logic* of the scheme; we do not claim to resolve RH.

Preliminaries

Fourier: $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-ix\xi} dx$, $\phi(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\phi}(\xi) e^{ix\xi} d\xi$. Let $\mathcal{PW}$ be even, smooth, band-limited functions ($\operatorname{supp} \widehat{\phi} \subset [-A, A]$).

T1′ — Explicit formula (Guinand–Weil)

$$\sum_{\rho} \widehat{\phi}(\gamma_\rho) = \frac{1}{2\pi} \int K(t) \widehat{\phi}(t)\, dt - \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \big(\phi(\log n) + \phi(-\log n)\big), \qquad K(t) = 2\log|t| + O(|t|^{-2}).$$

T2 — Spectral framing (Gram + Gershgorin/Schur)

Grid $\Xi_A = \{\xi_{p,k} = k \log p \leq A\}$, weights $w_{p,k} = \log p / p^{k/2}$, Gram $G^{(w)}_{\alpha\beta} = \sqrt{w_\alpha}\, \widehat{\phi}(\xi_\alpha - \xi_\beta)\, \sqrt{w_\beta}$, $R = G^{(w)} - I$. If $S := \sup_\alpha \sum_{\beta \neq \alpha} |R_{\alpha\beta}| < 1$ then $\lambda_{\min}(G^{(w)}) \geq 1 - S > 0$. Locality ($|\xi_\alpha - \xi_\beta| \leq A$) + p-adic decay $\Rightarrow S < 1$ for large A.

T3 — Wave-packet localization (contradiction in this scheme)

Assume an off-line zero $\rho_0 = \beta_0 + i\gamma_0$ ($\beta_0 \neq \frac{1}{2}$, $0 < \gamma_0 < A$). Build $\widehat{\phi}_T(\gamma) = \frac{1}{2}\big(b(\sqrt{T}(\gamma - \gamma_0)) + b(\sqrt{T}(\gamma + \gamma_0))\big)$, $\|\phi_T\|_2 = 1$. Then: Archimedean term $\to 0$ (DCT); prime-side $\to 0$ (IBP decay, finitely many $\xi_{p,k} \leq A$); zero-side $\to 1$ (frame capture). Plug into T1′ $\Rightarrow$ contradiction (in this framing).

T4 — Cross-positivity (equality criterion)

$K_\phi(\rho, \rho') = \widehat{\phi}(\gamma - \gamma')$ is positive; so is the prime-side quadratic form. Cauchy–Schwarz yields a cross-inequality. Equality for *all* fine tapers forces critical symmetry.

We prove the ***scheme*** *and its key bounds; we do not claim to settle RH.*